The Mirror Branch Braid Atlas: A Parameter-Free Dark Sector from the Universal Generative Principle

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Abstract

The Universal Generative Principle (UGP) derives the Standard Model spectrum from a parameter-free arithmetic sieve at ridge $n = 10$. The prime-lock mechanism that selects the SM lepton seed $(1, 73, 823)$ simultaneously forces a mirror-branch seed $(1, 73, 2137)$, predicting a complete dark sector with no free parameters.

We derive the Mirror Branch Braid Atlas—the dark sector particle classification table. The electric charge $Q = 0$ for all dark particles is derived from Braid Atlas topology, not assumed: the mirror $\mathbb{Z}_2$ involution is an internal GTE arithmetic symmetry, not an SM gauge transformation. The dark sector gauge group is $\mathrm{SU}(3)_{\mathrm{dark}}$ only—no dark $\mathrm{SU}(2)_L$ exists. Three generations of free dark leptons at 0.54 MeV, 24.5 MeV, and 3.60 GeV coexist with confined dark quarks near $\Lambda_{\mathrm{dark}} \sim 200\,\mathrm{MeV}$.

We establish a new structural connection: $c(W) = b_2(\mathrm{RHN}) = q_1$ in both branches—the first cross-sector structural unification in UGP Physics. The core results are machine-certified in Lean 4 (`MirrorWindingNumber.lean`, `EWBosonRHNConnection.lean`, `DarkBraidAtlas.lean`; zero sorry, zero custom axioms).

The dark neutrino mass-squared ratio $R_{\mathrm{dark}} = 0.208$ is $7\times$ larger than the SM ratio (**A/D**: MDL doublet-pairing derivation of $b_3' = 37$). The FIMP mechanism overproduces dark matter by 4.84×10^6 at $\lambda_s \sim 10^{-6}$; the leading resolution is asymmetric dark matter with $\eta_\chi \approx (2/7)\eta_{B+L,\mathrm{pre}}$, where $2/7 = (\mathbb{Z}_7$ charge of dark baryon$)/(\mathbb{Z}_7$ group order$)$. The most accessible prediction is GTE-P7 at 211.9 MeV—a Tier 1 target at Belle II. The baryogenesis derivation and dark confinement scale are identified as primary open problems.

Contents

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Claim-strength taxonomy

$$\mathbf{A}_{\text{Lean}} > \mathbf{A}_{\text{MDL}} > \mathbf{A/D} > \mathbf{B} > \mathbf{D}$$

$\mathbf{A}_{\text{Lean}}$**:** machine-checked exact arithmetic in Lean 4, zero `sorry`.
$\mathbf{A}_{\text{MDL}}$**:** MDL-unique within a declared expression class.
A/D: computationally confirmed; a physics bridge remains.
B: calibrated reproduction.
D: exploratory / frontier.

1 Introduction and Mirror Branch Structure

1.1 The UGP Dark Sector Prediction

The Universal Generative Principle (UGP) is a parameter-free framework that derives the Standard Model particle spectrum from an arithmetic sieve operating at ridge $n = 10$ [11, 17]. The sieve selects a unique lepton seed $(1, 73, 823)$ via three simultaneous constraints: prime-lock (both c_1 and its mirror must be prime), mirror-duality (a $b_2 \leftrightarrow q_2$ exchange symmetry of the cascade), and minimal description length (MDL) minimality. The resulting Braid Atlas assigns particle types to GTE orbits and derives Standard Model quantum numbers from arithmetic structure [12].

A central result of this paper is that the same prime-lock mechanism that selects the SM seed simultaneously *requires* a mirror branch—a second arithmetic solution at ridge $n = 10$ with $b_2 \leftrightarrow q_2$ exchanged: seed $(1, 73, 2137)$. This mirror branch is not optional; it is forced by the same sieve that forces the SM. The two seeds form an arithmetic pair: neither can be accepted without the other.

The mirror branch predicts a complete dark sector with specific, parameter-free particle masses and quantum numbers. Unlike conventional dark sector models where masses are free parameters, the UGP dark sector masses emerge from the same arithmetic that determines the SM masses—with zero additional freedom. This mirror-branch dark sector (defined by the $\mathbb{Z}_2$ involution; $Q = 0$ for all dark particles) is conceptually distinct from the $\mathbb{Z}_7$-winding dark sector discussed in Paper P28 [13], where the dark windings $\{1, 5\}$ are identified with SM anti-quarks carrying $Q \neq 0$. The two constructions are complementary: P28 identifies the dark-charge structure within the visible-sector $\mathbb{Z}_7$ orbit, while the present paper constructs the independent mirror-branch sector via the GTE arithmetic sieve.

The claim that the mirror branch corresponds to a *physical* dark sector—rather than being a mathematical redundancy of the GTE arithmetic—rests on three independent pillars: (1) the mirror branch produces distinct, parameter-free particle masses (0.54 MeV, 24.5 MeV, 3604 MeV, 211.9 MeV) that are independently testable at existing and planned detectors; (2) the GTE-P7 prediction at 211.9 MeV constitutes a Tier 1 Belle II target, providing a decisive experimental test within the decade; (3) if no mirror-branch particles are found at Belle II with the predicted properties and cross-sections, this result falsifies the dark sector identification, making the framework empirically distinguishable from the no-dark-sector alternative.

1.2 Contributions of This Paper

This paper establishes the following results:

1. The Mirror Braid Atlas: particle classification for all dark sector states, with quantum number assignments derived (not assumed) from GTE arithmetic.
2. $Q_{\text{EM}} = 0$ for all dark sector particles, derived from the Braid Atlas winding number theorem (not assumed), machine-certified in Lean 4 with zero custom axioms.

3. Two dark subsectors: free dark singlet leptons (3 generations, FIMP sector) and confined dark quarks forming dark hadrons at scale Λ_{dark}.
4. The EW Boson–RHN structural connection: $c(W) = b_2(\text{RHN}) = q_1$ in both branches (first cross-sector structural unification; Lean-certified).
5. The dark neutrino mass-squared ratio $R_{\text{dark}} = 0.208$, compressed 7× relative to the SM (**A/D**: MDL doublet-pairing derivation).
6. The relic density overproduction factor 4.84×10^6 at $\lambda_s \sim 10^{-6}$, and the leading asymmetric dark matter resolution with $\eta_\chi \approx (2/7)\eta_{B+L}$.
7. Experimental predictions and detection prospects, including the two-peak signature at 0.54 MeV and 24.5 MeV.

1.3 Comparison with Existing Dark Sector Models

The UGP dark sector differs from all existing dark sector proposals in several decisive respects. Table 1 summarizes the key distinctions.

Table 1: Comparison of the UGP mirror branch with representative dark sector models. "$Q = 0$" refers to the electric charge of the lightest dark state.

Model	$Q = 0$ mechanism	Mass spectrum	Dark gauge group
WIMP	Assumed	Free (GeV–TeV)	Free
Axion	By construction	Free (μeV–meV)	$\text{U}(1)_{\text{PQ}}$
Sterile neutrino	Assumed	Free (keV–GeV)	None
Dark photon	Assumed	Free, ε free	Dark $\text{U}(1)_D$
Mirror matter [3]	$Q = -1$ (excluded)	Free	Copy of SM
UGP mirror branch	Derived from braid topology	Fixed: 0.54, 24.5, 3600 MeV	$\text{SU}(3)_{\text{dark}}$ only

The decisive distinction is that in UGP, $Q = 0$ is a *theorem* derived from the braid winding number, not an assumption. Mirror matter models (Foot, Lew, Volkas [3]) assign $Q = -1$ to the dark electron by analogy with the SM; this is excluded by BBN and CMB constraints because mirror-charged particles would have thermalized with mirror photons and left a detectable radiation imprint. The UGP mirror branch avoids this because the $b_2 \leftrightarrow q_2$ swap maps GTE arithmetic invariants, not SM gauge charges: the dark particles are SM gauge singlets, giving $Q = 0$ as a structural consequence.

1.4 Lean Certificate Summary

All central structural claims have Lean 4 machine-checked proofs in the UGP Lean repository [18]. Table 2 summarizes the key certificates.

Table 2: Lean 4 certificates for central claims. All files are in the `UgpLean/BraidAtlas/` or `GTE/` directories of the UGP Lean repository. "Zero sorry" means the proof compiles with no admitted gaps; "zero axioms" means no custom axioms beyond Lean/Mathlib foundations.

Lean file	Key theorem(s)	Status
`MirrorWindingNumber.lean`	$W_g = 0$, $Q = 0$ for dark leptons (all gens); orbit distinguishability	Zero axioms
`DarkQuarkCharge.lean`	`dark_quark_charge_zero` (all gens)	Zero sorry, zero axioms
`DarkBraidAtlas.lean`	$b'_1 = 5$, $b'_2 = 29$, $q'_1 = 29$; $\mathrm{SU}(2)_L$ singlet structure	Zero sorry (inherits definitions from P17 canonical modules)
`EWBosonRHNConnection.lean`	$c(W) = b_2(\mathrm{RHN}) = q_1$ (both branches)	Zero sorry, zero axioms
`RHNGapTheorem.lean`	$19 = 11 + 8$, $37 = 29 + 8$ (arithmetic cert); MDL doublet-pairing structural derivation: $b_3 = q_2 - b_{1,\mathrm{RHN}}$ (**A/D**, §§6–7)	Zero sorry
`GTBGenerationPrimes.lean`	All 6 generation c_1 values prime; 3-generation structure	Zero sorry

The Lean certification distinguishes this work from all prior dark sector proposals: the structural claims are not merely argued but machine-verified against formal definitions. The structural gap theorem $b_3 = q_2 - b_{1,\mathrm{RHN}}$ has now been derived at **A/D** from the MDL doublet-pairing rule (§4.1); the derivation upgrades the dark neutrino hierarchy to a structural prediction conditional on the doublet-pairing principle. A full $\mathbf{A}_{\mathrm{Lean}}$ derivation from GTE axioms remains open.

1.5 Paper Structure

Section 2 derives the Mirror Braid Atlas and establishes the dark particle spectrum, including $Q = 0$ for dark leptons and dark quarks. Section 3 proves the EW Boson–RHN structural connection and its mirror-branch extension. Section 4 derives the dark neutrino mass hierarchy. Section 5 presents experimental predictions: the two-peak MeV signature, dark confinement structure, relic density status, and detection prospects. Section 6 concludes. Appendix A provides the full Lean certificate inventory.

2 Mirror Branch Braid Atlas: Dark Sector Classification

2.1 Mirror Branch Structure

The GTE arithmetic sieve at ridge $n = 10$ has exactly two outputs satisfying all prime-lock constraints: the SM sector and the mirror sector. Both arise from the same prime-lock mechanism; neither can be accepted without the other.

The two branches are distinguished by the swap $b_2 \leftrightarrow q_2$, which exchanges two outputs of the same prime-lock computation. The arithmetic parameters are:

Parameter	SM branch	Mirror branch
b_2	42	24
q_2	24	42
c_1	823	2137
q_1	11	29

Both $c_1 = 823$ and $c_1' = 2137$ are prime (Lean-certified). The mirror seed satisfies $c_1' = b_2' \cdot q_2' + 20 = 24 \times 89 + 1 = 2137$ with residue $2137 \bmod 73 = 20$, matching the SM residue class. The pair is unique: `prime_lock_unique_survivor` in the UGP Lean 4 formalization library [18] certifies that no other pair at ridge $n = 10$ satisfies all constraints simultaneously.

The swap $b_2 \leftrightarrow q_2$ is an arithmetic symmetry of the GTE cascade integers. It is *not* an SM gauge transformation: mirror-branch particles carry no SM gauge charges [12]. This is the physical origin of the dark sector's electrical neutrality.

The branch-shift invariant is $\Delta q \equiv q_2 - q_1 = 24 - 11 = 13 = q_2' - q_1' = 42 - 29$. This spacing is preserved across branches and provides an internal consistency check on the arithmetic structure. The UGP-1 branch-constructor formula $q_1 = q_2 - g$ uses $g = u_g = 13 = c_H$ as its gap parameter (machine-certified: `ugp1_g_eq_cH`, `branch_spacing_is_ugp1_g_universal`; `GUTStructure.lean` §51). Together with the CA identity $T_{\text{ether}} = c_H + 1 = 14$ (`ether_period_cH_structural`, `GUTStructure.lean` §40), this yields the cross-level structural identity

$$T_{\text{ether}} = \Delta q + 1 = 14,$$

machine-certified as `ether_mirror_structural_identity` (`GUTStructure.lean` §51, zero `sorry`): the Rule 110 ether period equals the mirror-branch arithmetic spacing plus one.

2.2 Braid Topology of Mirror Particles

Mirror-branch particles occupy the same braid orbit structure as their SM analogues. Dark leptons correspond to the single-strand ($a_1 = 1$) orbit class, analogous to SM charged leptons and neutrinos. Dark quarks are single-strand under the dark SU(3), though SM $\mathrm{SU}(3)_c$ acts trivially on them (dark quarks carry dark color, not SM color).

The mirror orbit ($b_2' = 24, q_2' = 42, c_1' = 2137$) is arithmetically distinct from the SM lepton orbit ($b_2 = 42, q_2 = 24, c_1 = 823$): same residue class $m_1 = 20$ (same GTE orbit family), different representatives ($c_1' = 2137 > 823 = c_1$), and different first quotients ($q_1' = 29 \neq 11 = q_1$). This is certified by `MirrorWindingNumber.lean` (zero axioms): the orbit distinguishability theorems `mirror_c1_ne_lepton_c1` and `mirror_q1_ne_canonical_sm_q1` confirm the arithmetic separation.

2.3 Dark Singlet Leptons: Three Generations, $Q = 0$ Derived

The mirror branch generates three generations of dark singlet leptons. Their quantum numbers and masses are given in Table 3.

Table 3: Dark singlet lepton spectrum. Masses are parameter-free predictions of the GTE cascade arithmetic applied to the mirror branch (not Lean-certified at the mass level; see text). The charge $Q_{\mathrm{EM}} = 0$ is a Lean-certified theorem (zero axioms).

Generation	Label	Mass	Q_{EM}	Primary search venue
G1	χ_1	0.5406 MeV	0	Sub-MeV dark matter (Higgs portal)
G2	χ_2	24.47 MeV	0	MeV-scale dark matter (Higgs portal)
G3	χ_3	3604.68 MeV	0	LHC Higgs invisible / mono-jet (λ_s suppressed)

Derivation of $Q = 0$ (not assumed). The mirror orbit has $Y_{\mathrm{mirror}} = 0$ and $T_{3,\mathrm{mirror}} = 0$ (no SM hypercharge, no SM weak isospin) by the universal mirror-duality statement of P17 [12]: the $b_2 \leftrightarrow q_2$ swap is an internal GTE arithmetic operation, not an SM gauge transformation, so mirror-branch particles carry no SM gauge quantum numbers. The standard Gell-Mann–Nishijima formula $Q = T_3 + Y/2$ then gives $Q = 0 + 0 = 0$. In the UGP GTE convention, the *winding number* $W_g \equiv N_c \times Q$ (an integer multiple of the charge by the colour rank $N_c = 3$) satisfies:

$$W_g = N_c \times (T_3 + Y/2) = 3 \times 0 = 0 \quad \Longrightarrow \quad Q = W_g/N_c = 0. \tag{1}$$

This is a theorem from definitions, not an axiom. The Lean 4 proof is in `MirrorWindingNumber.lean` (zero custom axioms in this proof chain): the theorem `mirror_lepton_q_em_zero` proves $Q = 0$ for all three generations simultaneously via `dark_lepton_q_em_zero_all_gens`.

Why this differs from mirror matter models. Mirror matter proposals [3] assign $Q = -1$ to the dark electron because they map the SM electron—including its SM gauge charges—to a mirror partner. In the UGP mirror branch, the $b_2 \leftrightarrow q_2$ swap operates on GTE arithmetic labels, not on SM gauge representations. The dark particles are SM gauge singlets from the outset. There is no mirror electric charge, no mirror photon, and no mirror electromagnetic coupling. The charge $Q = 0$ follows from $Y_{\mathrm{mirror}} = 0$ as a structural consequence of the GTE prime-lock.

Dark lepton mass hierarchy. The three-generation mass hierarchy satisfies:

$$\frac{G2}{G1} = \frac{24.47}{0.5406} \approx 45.3 \qquad \text{vs. SM} \quad \frac{m_\mu}{m_e} \approx 206.8, \tag{2}$$

$$\frac{G3}{G2} = \frac{3604.68}{24.47} \approx 147.3 \qquad \text{vs. SM} \quad \frac{m_\tau}{m_\mu} \approx 16.8. \tag{3}$$

The G2/G1 ratio is compressed by a factor of approximately 4.6 relative to the SM charged lepton hierarchy. This compression is not tuned; it is determined by the mirror branch GTE values ($b_2' = 24$ vs. SM $b_2 = 42$). The G3/G2 ratio is *enhanced* relative to SM τ/μ, producing an asymmetric overall hierarchy that is distinctive. Both ratios are falsifiable predictions.

2.4 Dark Quarks: $Q = 0$, Dark SU(3) Colour, Confinement

Dark quarks carry $Q_{\mathrm{EM}} = 0$. This follows from the same universal $Y_{\mathrm{mirror}} = 0$ argument as for dark leptons. The key distinction is that dark quarks *do* carry a dark SU(3) colour charge (distinct from SM $\mathrm{SU}(3)_c$), but this internal dark charge does not contribute to the Gell-Mann–Nishijima formula, which is defined with respect to SM quantum numbers.

The Lean 4 proof is in `DarkQuarkCharge.lean` (zero sorry, zero axioms), which contains three key theorems: (i) `dark_quark_charge_zero`, which proves $W_g^{\mathrm{mirror}} = 0$ from the mirror winding number; (ii) `dark_quark_charge_zero_all_gens`, which extends the result to all generations; and (iii) a composite machine-checked certificate that the quark charge result holds across all mirror-branch generations simultaneously. The proof is independent of the lepton case—it establishes $Q = 0$ for the dark quark sector separately.

Table 4 contrasts SM quarks with their dark counterparts.

Table 4: SM quarks vs. dark quarks: quantum number comparison. Dark quarks carry dark SU(3) colour (not SM colour) and have $Q_{\mathrm{EM}} = 0$ (Lean-certified).

Particle	W_g	Q_{EM}	Colour
SM up quark	$+2$	$+2/3$	SM $\mathrm{SU}(3)_c$
SM down quark	-1	$-1/3$	SM $\mathrm{SU}(3)_c$
Dark up quark G1	0	0	Dark SU(3)
Dark down quark G1	0	0	Dark SU(3)

The dark quark G1 masses are 0.57 MeV (dark up) and 17.30 MeV (dark down), derived from the mirror-branch Permutation Principle (preliminary, pending verification from the locked GTE map). Dark G2 and G3 quark masses are not yet computable from current GTE results. Since dark quarks carry dark SU(3) colour, they confine at the dark confinement scale Λ_{dark} into dark hadrons (dark pions, dark protons); see Section 5.2.

2.5 Two-Component Dark Sector

The mirror branch produces a two-component dark sector:

Component 1: Free dark singlet leptons (FIMP sector). Dark singlet leptons (χ_1, χ_2, χ_3) are electrically neutral (Lean-certified), dark SU(3) singlets (they are not quarks and do not confine), and produced via freeze-in (FIMP mechanism) through the Higgs portal. The FIMP mechanism—introduced in [4, 7]—is specifically designed for very small couplings ($\lambda_s \sim 10^{-6}$, two-loop suppressed), which is precisely the UGP portal coupling.

Component 2: Confined dark hadrons (dark strong sector). Dark quarks carry dark SU(3) colour and confine at Λ_{dark} into dark hadrons. The confinement structure is discussed in Section 5.2.

2.6 The Zero-Parameter Distinction

Every existing dark sector model introduces at least one free parameter: WIMP mass, axion decay constant, sterile neutrino mixing angle, dark photon kinetic mixing. The UGP mirror branch has zero free parameters (Table 5).

Table 5: Free parameter count in representative dark sector models vs. UGP.

Model	**Free params**	$Q = 0$ **status**	**DM mass range**
WIMP	$\geq$ 2 (mass + coupling)	Assumed	GeV–TeV
Axion	≥ 2 (mass + f_a)	By construction	μeV–meV
Sterile ν	$\geq$ 2 (mass + mixing)	Assumed	keV–GeV
Dark photon	≥ 2 (mass + ε)	Assumed	MeV–GeV
Mirror matter [3]	≥ 1 (scale)	$Q = -1$ (excluded)	Broad
UGP mirror branch	**0**	Derived (Lean 4, zero axioms)	0.54, 24.5, 3600 MeV

The UGP dark sector is not a model but a prediction: the arithmetic sieve that selects the SM necessarily produces this dark sector with the same number of free parameters—zero.

2.7 Dark-Sector Portal: $\mathbb{Z}_7$ Complement Selection Rule

The $\mathbb{Z}_7$ additive complement structure forces a specific dark-sector portal. The two dark-sector winding classes $\{w = 1, w = 5\}$ pair exclusively with SM quarks (d-quark $w = 6$, u-quark $w = 2$) as additive $\mathbb{Z}_7$ complements: $1 + 6 \equiv 0 \pmod 7$ and $5 + 2 \equiv 0 \pmod 7$ (CatA, arithmetic). Leptons, W bosons, and photons have no dark-sector additive complements: $W^+(w = 3)$ and $e^-(w = 4)$ are each other's complements, both in the SM sector. This is a sharp, parameter-free prediction: GTE dark matter production couples exclusively to quarks (nucleons), not to leptons or gauge bosons. Models with leptophilic dark matter are excluded at the CA arithmetic level. (Script: `vacuum_pair_production.py`, CatA; Lean target: `decide` proof via enumeration.)

2.8 $\mathbb{Z}_2$ Mirror-Dual Generating Triple and Dark GTE Master Formula

The $\mathbb{Z}_2$ duality $v \mapsto 7 - v$ on $\mathbb{Z}_7$ winding numbers maps the SM branch to the dark-sector branch identified above. Under this duality the SM W^+ boson ($\mathbb{Z}_7$ winding $w = 3$) maps to the dark W'^+ with winding $w' = 7 - 3 = 4$. The GTE identification $N_{\text{gen}} = W(W^+)$ gives $N_{\text{gen}}^{\text{SM}} = 3$ from the SM W^+ winding $W(W^+) = 3$. This identification applies equally to the mirror branch: the dark analogue $W'(W'^+) = 4$, giving $N_{\text{gen}}^{\text{dark}} = 4$ under the same structural identification (**A/D**).

With $N_{\text{fam}}^{\text{dark}} = 5$ (the $\mathbb{Z}_5$ ring structure is invariant under the $\mathbb{Z}_7$ duality; **A/D**, N_{fam} ambiguity noted below), the dark Higgs branch capacity is

$$c_H^{\text{dark}} \;=\; N_{\text{gen}}^{\text{dark}} + 2N_{\text{fam}}^{\text{dark}} \;=\; 4 + 10 \;=\; 14. \tag{4}$$

The dark sector GTE master formula then predicts:

$$\sin^2\theta_W^{\text{dark}} \;=\; \frac{N_{\text{gen}}^{\text{dark}}}{c_H^{\text{dark}}} \;=\; \frac{4}{14} \;=\; \frac{2}{7} \quad (\mathbf{A/D}), \tag{5}$$

$$\sin^2\theta_W^{\text{dark}}(M_{\text{GUT}}) \;=\; \frac{N_{\text{gen}}^{\text{dark}}}{2^{N_{\text{gen}}^{\text{dark}}}} \;=\; \frac{4}{16} \;=\; \frac{1}{4} \quad (\mathbf{A/D}), \tag{6}$$

$$\lambda^{\text{dark}} \;=\; \sin^2\theta_W^{\text{dark}}(M_{\text{GUT}}) \times \frac{N_{\text{gen}}^{\text{dark}}}{N_{\text{fam}}^{\text{dark}}} \;=\; \frac{1}{4}\cdot\frac{4}{5} \;=\; \frac{1}{5} \quad (\mathbf{A/D}). \tag{7}$$

These are parameter-free predictions of the mirror-branch GTE arithmetic.

Two internal consistency checks support the dark GTE formula. First, the ratio of dark to SM step-b values is $b_2'/b_2 = 24/42 = 4/7$, which equals the Rule 110 ether density (8 out of 14 ether cells carry $\mathbb{Z}_7 = 1$, i.e., $8/14 = 4/7$; CatA). Second, the dark branch cascade value $b_2' = 24$ satisfies $24 \bmod 7 = 3$, matching the $\mathbb{Z}_7$ winding of the SM W^+ boson — the $\mathbb{Z}_2$ duality is reflected at the mod-7 level in the GTE seed parameters (CatA, arithmetic).

Remark 2.1 ($N_{\text{fam}}^{\text{dark}}$ ambiguity and certification status)**.** *The derivation above assumes $N_{\text{fam}}^{\text{dark}} = 5$, matching the SM family-ring size. An alternative assignment $N_{\text{fam}}^{\text{dark}} = 12$ (from the dark sector cascade dimension count) would give $c_H^{\text{dark}} = 28$ and $\sin^2\theta_W^{\text{dark}} = 1/7$. The $N_{\text{fam}}^{\text{dark}} = 5$ assignment is preferred by the $\mathbb{Z}_5$ ring universality argument but is not yet Lean-certified; the prediction is therefore **A/D**. Script: `dark_sector_master_formula.py`.*

3 EW Boson–RHN Structural Connection

3.1 The Connection in the SM Branch

Computing separately from the GTE cascade at ridge $n = 10$ the c-value of the W boson and the b_2 parameter of the right-handed neutrino sector—two outputs from different stages of the cascade that were not previously recognized as related—we establish:

$$c(W) = b_2(\text{RHN}) = q_1 = 11. \tag{8}$$

The common value is $q_1 = 11$, the first GTE quotient at ridge $n = 10$. The identity $c(W) = q_1$ follows from the W boson c-value computation in the SM branch [12]; the identity $b_2(\text{RHN}) = q_1$ follows from the RHN cascade step [16]. That both equal the same q_1 is structural: both are outputs of the GTE prime-lock at the same ridge, indexed by the first quotient.

The Lean 4 certificate is `EWBosonRHNConnection.lean` (zero sorry, zero axioms—the strongest possible certification). The composite theorem `ew_rhn_connection_both_branches` states:

```
c_W = b2_RHN ∧ c_W = q1
```

This connection was not stated or proved in the preceding UGP papers. P29 is the first to identify and formally certify this structural link.

3.2 Extension to the Mirror Branch

The connection extends exactly to the mirror branch with mirrored values. Since $q_1' = 29$ for the mirror branch (see Section 2.1), and $b_2'(\text{RHN}) = 29$ (Lean-certified via `gteQuotient` in `DarkBraidAtlas.lean`), we have:

$$q_1'(\text{mirror}) = b_2'(\text{RHN}) = 29. \tag{9}$$

The branch shift is preserved: $q_1'(\text{mirror}) - q_1(\text{SM}) = 29 - 11 = 18$. The branch-invariant spacing $\Delta q \equiv q_2 - q_1 = 13$ is the same in both branches (SM: $24 - 11 = 13$; mirror: $42 - 29 = 13$).

It is important to note that $q_1' = 29$ in the mirror branch is a GTE arithmetic invariant—it is *not* the c-value of a dark W' boson. The dark sector has no electroweak symmetry (Section 3.4), so no dark W' exists. The value 29 labels the arithmetic root of the mirror branch, and its coincidence with $b_2'(\text{RHN})$ is the structural connection being reported.

3.3 Physical Interpretation

The EW gauge sector (W boson) and the neutrino mass sector (right-handed neutrino) share the same arithmetic root q_1. This is the first cross-sector structural unification in the UGP Physics framework.

In ordinary quantum field theory, there is no expectation that gauge boson parameters and fermion mass parameters share a common value—they enter from independent Lagrangian terms and are treated as unrelated free parameters. In the GTE framework, both W boson c-values and RHN b-values are outputs of the same cascade at the same ridge. Their shared arithmetic root q_1 is a consequence of GTE architecture.

Conjecture 3.1 (EW–RHN cascade identity)**.** *The structural identity $c(W) = b_2(\mathrm{RHN}) = q_1 = 11$ (and its mirror analog $q_1' = 29$) suggests that the EW cascade parameter $c_H = 13$ plays a dual role—as the Higgs cascade depth in the electroweak sector and as the first cascade quotient connecting the SM and RHN branches. Whether this arithmetic coincidence has a deeper structural explanation from GTE minimality (analogous to the derivation of $b_H = 3$) remains open.*

3.4 No Dark Electroweak Symmetry

A corollary of the $Q = 0$ derivation is that the dark sector has *no* electroweak symmetry. Since $T_{3,\mathrm{mirror}} = Y_{\mathrm{mirror}} = 0$ for all mirror-branch particles (Lean-certified in `DarkBraidAtlas.lean`), all dark particles are $\mathrm{SU}(2)_L$ singlets. There is no dark $\mathrm{SU}(2)_L$, no dark $\mathrm{U}(1)_Y$, and no dark Higgs mechanism. The dark sector gauge group is:

$$G_{\mathrm{dark}} = \mathrm{SU}(3)_{\mathrm{dark}} \quad \text{(only)}. \tag{10}$$

This rules out a dark W', dark Z', or dark Higgs boson from an electroweak mechanism. It also rules out kinetic mixing between a dark photon and the SM hypercharge boson: since no dark $\mathrm{U}(1)_D$ exists, the kinetic mixing parameter $\varepsilon = 0$ exactly by structure (not by fine-tuning). This is a falsifiable structural prediction: a discovery of a dark photon signal would falsify the UGP dark sector at the highest confidence level.

4 Dark Sector Neutrino Mass Structure

4.1 Mirror RHN b-Values

The mirror branch GTE cascade produces three right-handed neutrino b-values, which govern the dark neutrino mass hierarchy via the same mass formula established in P21 [16]. The values and their certification status are:

Parameter	Value	Certificate	Notes
$b_1'(\mathrm{RHN})$	5	Branch-invariant	$b_1 = 2N_c - 1 = 5$; holds for both branches
$b_2'(\mathrm{RHN})$	29	Lean-certified	`dark_b2_rhn_eq_29` in `DarkBraidAtlas.lean`
$b_3'(\mathrm{RHN})$	37	**A/D** (MDL doublet-pairing)	$b_3' = q_2' - b_1 = 42 - 5 = 37$; structural derivation from MDL rule (§4.1)

The branch invariance of $b_1 = 5$ follows from the N_c formula: $b_1 = 2N_c - 1 = 5$ regardless of which branch is selected, because $N_c = 3$ is an SM structural constant shared by both branches.

The value $b_3' = 37$ is now established at **A/D** via the MDL doublet-pairing rule. The derivation chain is:

1. *EW-RHN connection* ($\mathbf{A}_{\text{Lean}}$, `EWBosonRHNConnection.lean`): $b_2'(\text{RHN}) = q_2' - u_g = 42 - 13 = 29$, where $u_g = 13$ is the GTE quotient gap.
2. *Key structural identity* ($\mathbf{A}_{\text{Lean}}$, `RHNGapTheorem.lean` §6): $u_g - b_1 = N_c^2 - 1 = 8$, connecting the Higgs gap $u_g = 13$ and the RHN seed $b_1 = (N_c^2+1)/2 = 5$ via the SU(3) adjoint dimension.
3. *MDL doublet-pairing rule* (**A/D**): the RHN triple satisfies $b_1 + b_3 = q_2$ (doublet sum rule), so $b_3' = q_2' - b_1 = 42 - 5 = 37$.
4. *Algebraic chain*: $b_3' = b_2' + (u_g - b_1) = 29 + 8 = 37$, showing the numerical observation $b_3 = b_2 + (N_c^2 - 1)$ is structurally forced.

All arithmetic is certified by `RHNGapTheorem.lean` (zero sorry, §6–7). Upgrading to full $\mathbf{A}_{\text{Lean}}$ requires proving the doublet sum rule $b_1 + b_3 = q_2$ from GTE axioms—an open problem for future work.

4.2 Dark Neutrino Mass Hierarchy

Given the mirror RHN b-values, the dark neutrino mass-squared ratio is:

$$R_{\text{dark}} = \frac{{b_2'}^{58/9} - {b_1'}^{58/9}}{{b_3'}^{58/9} - {b_1'}^{58/9}} = \frac{29^{58/9} - 5^{58/9}}{37^{58/9} - 5^{58/9}} = 0.2080. \tag{11}$$

This formula is the same as the one derived for the SM in P21 [16], applied to the mirror branch b-values. The exponent 58/9 emerges from the GTE cascade arithmetic and is treated here as branch-invariant—the same structural formula that applies to the SM branch is applied to the mirror branch. This is an assumption, not a derived result: a first-principles derivation of the exponent for the mirror branch separately remains an open item. The SM ratio, $R_{\text{SM}} = 0.02936$, is Lean-certified in P21 and matches NuFIT 6.0 [2] to within 0.16σ.

Quantity	Value	Certificate	Notes
R_{SM}	0.02936	Lean-certified (P21)	Matches NuFIT 6.0 at 0.16σ
R_{dark}	0.2080	**A/D** (MDL doublet-pairing)	See note below
$R_{\text{dark}}/R_{\text{SM}}$	≈ 7.09	**A/D**	Dark hierarchy $7\times$ larger

The dark neutrino hierarchy follows normal ordering ($b_1' < b_2' < b_3'$), the same ordering as the SM. The hierarchy is compressed at the G1/G2 step (where SM and dark differ most) and enhanced at the G2/G3 step.

Note: $R_{\text{dark}} = 0.2080$ now rests on a **A/D** structural derivation of $b_3' = q_2' - b_1' = 37$ from the MDL doublet-pairing rule (chain: EW-RHN connection + key identity $u_g - b_1 = N_c^2 - 1$ + doublet sum rule; all arithmetic certified in `RHNGapTheorem.lean` §6–7, zero sorry). This The value $b_3' = 37$ is established at **A/D** from the MDL doublet-pairing rule. The full $\mathbf{A}_{\text{Lean}}$ derivation—proving the doublet sum rule from GTE axioms—remains open.

4.3 Physical Discussion

The compression factor $R_{\text{dark}}/R_{\text{SM}} \approx 7$ is an arithmetic consequence of the mirror branch values ($b_2'(\text{RHN}) = 29$ vs. SM $b_2(\text{RHN}) = 11$), not a tunable parameter. The structural gap theorem is now established at **A/D** (MDL doublet-pairing rule), so R_{dark} is a structural prediction at that grade. Full $\mathbf{A}_{\text{Lean}}$ certification—proving the doublet sum rule from GTE axioms—would remove the last qualification. This compressed hierarchy is a potentially falsifiable prediction for dark sector cosmology. Access to three dark neutrino mass-squared splittings with the ratio $R_{\text{dark}}/R_{\text{SM}} \approx 7$ would be strong evidence for the UGP mirror branch.

The dark neutrinos are sterile with respect to all SM gauge interactions. Their hierarchy is in principle accessible via dark sector precision cosmology (contributions to N_{eff} or the dark radiation spectrum) or via future experiments probing dark radiation at sufficient energy resolution.

Conjecture 4.1 (Mirror-branch seesaw analogue, **A/D**). *The compression factor $R_{\text{dark}}/R_{\text{SM}} \approx 7$ is the GTE mirror-branch analogue of the SM seesaw mechanism. In the SM, the Weinberg–Salam type-I seesaw suppresses light neutrino masses by coupling them to a high-scale right-handed mass M_R: $m_\nu \sim m_f^2/M_R$. In the GTE mirror branch, the doublet-pairing rule assigns $b_2' = 29$ to the mirror second-generation RHN, versus $b_2 = 11$ in the SM. The ratio $b_2'/b_2 = 29/11 \approx 2.6$ propagates via the GTE mass formula to give the mass-squared ratio compression $R_{\text{dark}}/R_{\text{SM}} \approx 7$ (structurally derived at **A/D**; see §4.2). The structural gap theorem ($b_3' = q_2' - b_1'$, **A/D**) plays the role of the seesaw scale separation: it fixes b_3' relative to the EW–RHN connection, determining the hierarchy compression without a free parameter. Full $\boldsymbol{A}_{\text{Lean}}$ certification of this analogy awaits proof of the MDL doublet sum rule from GTE axioms.*

5 Experimental Predictions and Detection Prospects

5.1 Prediction Table

Table 6 summarizes all experimentally accessible dark states predicted by the UGP mirror branch.

Table 6: UGP dark sector predictions. All masses are parameter-free; none are fitted. Cert. status uses the claim-strength taxonomy: "Lean-cert" = $\mathbf{A}_{\text{Lean}}$ (machine-certified in Lean 4, zero sorry); "GTE arith" = **A** (derivation from GTE arithmetic, not Lean-certified at mass level); **A/D** = computationally confirmed from a named GTE structural principle; **D** = exploratory. $Q_{\text{EM}} = 0$ for all dark states is Lean-certified (zero axioms).

State	**Mass**	Q_{EM}	**Cert. status**	**Primary search venue**
GTE-P7	211.9 MeV	0	Lean-cert (P02)	BaBar/LHCb archival dimuon; Belle II Tier 1
Dark lepton G1 (χ_1)	0.5406 MeV	0	GTE arith	Sub-MeV DM; CMB spectral distortion (Higgs portal; below current sensitivity)
Dark lepton G2 (χ_2)	24.47 MeV	0	GTE arith	MeV-scale Higgs-portal DM; future dedicated search (below current sensitivity)
Dark lepton G3 (χ_3)	3604.68 MeV	0	GTE arith	LHC Higgs invisible width / mono-jet (λ_s suppressed; below current sensitivity)
Dark photon A'	**Does not exist**	—	Lean-cert	NA64/BaBar/Belle II null = consistent
$G2/G1$ mass ratio	45.4	—	GTE arith	Same experiment covering 0.5–25 MeV
R_{dark} (dark ν)	0.2080	—	**A/D** (MDL doublet-pairing)	Dark radiation precision cosmology
Λ_{dark}	~ 200 MeV–1.7 GeV	—	α_s Lean-cert; N_f unknown	Dark hadron searches

5.2 Dark Hadron Subsector and the Confinement Scale

The dark SU(3) coupling constant is *not* a free parameter. It is determined by the P01 Gauge Coupling Master Formula [11]:

$$g_G^2 = L_G \cdot D_G / 5^{\gamma_G}, \tag{12}$$

where L_G is the Weyl group order (fixed by group theory), D_G is the Vandermonde determinant squared (Elegant Kernel global MDL constant, not branch-specific), and $\gamma_G = 3$. For SU(3)$_{\text{dark}}$, the same inputs apply as for SM SU(3)$_c$:

$$\alpha_{s,\text{dark}}^{\text{bare}} = \frac{g_{3,\text{dark}}^2}{4\pi} = \frac{41{,}075{,}281}{27{,}648{,}000 \times 4\pi} = 0.11822 \quad \text{(same as SM)}. \tag{13}$$

This structural equality—$\alpha_{s,\text{dark}}^{\text{bare}} = \alpha_{s,\text{SM}}^{\text{bare}}$—follows from the shared Elegant Kernel and is not an assumption; it is derived from the same Vandermonde determinant that fixes the SM coupling. The formula is Lean-certified via `g3Sq_bare_eq` (zero sorry).

Given equal bare couplings, the dark confinement scale depends on the number of active dark quark flavors $N_{f,\text{dark}}$ at the running scale:

$$N_{f,\text{dark}} = 6 \Rightarrow \Lambda_{\text{dark}} \approx 210\,\text{MeV}, \tag{14}$$

$$N_{f,\text{dark}} = 2 \Rightarrow \Lambda_{\text{dark}} \approx 1.7\,\text{GeV}. \tag{15}$$

The value of $N_{f,\text{dark}}$ at the confinement scale is not yet determined from GTE arithmetic (dark G2/G3 quark masses are unknown); this introduces a range of [210 MeV, 1.7 GeV] for Λ_{dark}.

Self-consistency of the $N_{f,\text{dark}} = 2$ scenario. With $\alpha_{s,\text{dark}} = \alpha_{s,\text{SM}}(M_Z) = 0.118224$ (Lean-certified: `g3Sq_bare_eq`, zero sorry), the $N_{f,\text{dark}} = 2$ scenario is rigorously self-consistent: both G1 dark quarks ($m_{Q_1^d} \approx 0.57\,\text{MeV}$ and $m_{Q_2^d} \approx 17.3\,\text{MeV}$, from the GTE Permutation Principle cascade) are well below $\Lambda_{\text{dark}} \approx 1.7\,\text{GeV}$, confirming they are active at the confinement scale. The $N_{f,\text{dark}} \geq 3$ scenarios are blocked on the dark G2 quark mass, which requires the Permutation Principle cascade b-values for dark-sector G2 quarks (open problem). The two cases are summarised in Table 7.

Table 7: Λ_{dark} self-consistency status by $N_{f,\text{dark}}$. One-loop running from $\alpha_{s,\text{dark}}(M_Z) = 0.118224$ (Lean-certified).

$N_{f,\text{dark}}$	Λ_{dark}	Dark quarks below Λ_{dark}?	Status
2	$\approx 1.7\,\text{GeV}$	Yes: G1 quarks (0.57, 17.3 MeV)	**Self-consistent**
≥ 3	$\leq 1.7\,\text{GeV}$	Unknown: G2 quark mass not yet derived	Blocked
6	$\approx 210\,\text{MeV}$	Unknown	Blocked

Under the $N_{f,\text{dark}} = 6$ scenario, dark pion masses (PCAC estimate) are $\approx 11\,\text{MeV}$ (up-type) and $\approx 59\,\text{MeV}$ (down-type), and the dark proton mass is $\approx 630\,\text{MeV}$ ($\approx 3 \times \Lambda_{\text{dark}}$). These are preliminary order-of-magnitude estimates.

The GTE-P7 state at 211.9 MeV [15] falls within the dark baryon mass range under the $N_{f,\text{dark}} = 6$ scenario, providing an internal cross-check: GTE-P7 is consistent with being a dark baryon or dark hadron excited state above Λ_{dark}.

5.3 Current Experimental Status and the Dark Photon Non-Prediction

No dark state from the UGP mirror branch is currently excluded by experiment. The reason is the predicted Higgs portal coupling:

$$\lambda_s \sim 10^{-6} \quad \text{(two-loop suppressed)}, \tag{16}$$

which places all signal cross-sections 5–14 orders of magnitude below the current experimental sensitivity of BaBar, ATLAS/CMS, and the best sub-GeV dark matter searches.

The UGP dark sector is not in thermal equilibrium with the SM at any epoch. The FIMP production mechanism [4, 7]—freeze-in rather than freeze-out—means the dark sector never thermalized. It therefore does not contribute to the effective number of relativistic species N_{eff} in a way that would violate Planck 2018 constraints [1]. BBN constraints do not apply to the dark leptons ($Q = 0$, not in the SM thermal bath).

Stellar cooling and direct detection. The sub-MeV mass of χ_1 raises potential concerns about stellar energy-loss constraints. We address each in turn. *Horizontal branch stars* ($T_{\text{HB}} \sim 10\,\text{keV}$): since $m_{\chi_1}/T_{\text{HB}} \approx 54$, thermal production is Boltzmann-suppressed by $e^{-54} \approx 3 \times 10^{-24}$; horizontal branch bounds do not apply. *SN1987A* ($T_{\text{SN}} \sim 30\,\text{MeV}$): χ_1 is relativistic in the supernova core, but production via Higgs exchange is suppressed by $\lambda_s^2/(m_h^4)$. Translating to an effective Higgs mixing angle, $\sin^2\theta \approx (\lambda_s\, v_H/m_h^2)^2 \approx 2.5 \times 10^{-16}$ for $\lambda_s \sim 10^{-6}$, which lies 4×10^5 times below the SN1987A bound $\sin^2\theta < 10^{-10}$ [6]. The effective χ_1–nucleon coupling is $g_{\chi N} = \lambda_s f_N m_N/m_h^2 \approx 1.8 \times 10^{-11}$, giving an energy-loss rate negligible compared to the SN1987A constraint. *CMB spectral distortion*: the only kinematically accessible annihilation channel is $\chi_1\bar{\chi}_1 \to e^+e^-$ (barely open: $2m_{\chi_1} - 2m_e \approx 0.059\,\text{MeV}$); the cross-section at the FIMP coupling, $\langle\sigma v\rangle \approx 6 \times 10^{-67}\,\text{cm}^2$, is 40 orders of magnitude below the CMB spectral distortion bound [10]. *Direct detection*: the spin-independent χ_1–nucleon cross-section via Higgs exchange is $\sigma_{\text{SI}}(\chi_1\text{-}N) = 9.76 \times 10^{-61}\,\text{cm}^2$ (**A/D**; dark Yukawa $y_\chi = m_{\chi_1}/v_H$, Higgs exchange kernel); the FIMP coupling estimate ($\lambda_s \sim 10^{-6}$) gives $2 \times 10^{-61}\,\mathrm{cm}^2$, consistent within an order of magnitude. This is approximately 28–29 orders of magnitude below the best current experimental bound at $m \approx 0.54\,\mathrm{MeV}$ (CRESST-III: $\sigma < 3 \times 10^{-32}\,\mathrm{cm}^2$). Furthermore, the maximum nuclear recoil energy $E_R^{\max} = 2m_r^2 v^2/m_N \approx 1.7 \times 10^{-4}\,\mathrm{eV}$ at galactic velocity $v = 220\,\mathrm{km/s}$ lies seven orders of magnitude below the lowest current detector threshold (CRESST-III: 3.4 eV), placing χ_1 firmly in the ultra-feeble dark matter regime: conventional nuclear-recoil detectors are kinematically blind to χ_1. In all cases, the two-loop-suppressed coupling $\lambda_s \sim 10^{-6}$ is safe—no stellar cooling or direct detection bound is approached. For the asymmetric dark matter scenario discussed in Section 5.5, the coupling must have been large enough for χ_1 to reach chemical equilibrium with the SM at some temperature $T \gg m_{\chi_1}$; at $T \sim 1\,\mathrm{GeV}$ this requires $\lambda_s \gtrsim 3 \times 10^{-10}$, well within the allowed window $[3 \times 10^{-10}, 6 \times 10^{-4}]$ bounded above by SN1987A.

Dark photon non-prediction. A key structural prediction from the Mirror Braid Atlas is the *absence* of a dark photon. Since all mirror-branch particles have $Y_{\mathrm{mirror}} = T_{3,\mathrm{mirror}} = 0$ (Lean-certified in `DarkBraidAtlas.lean` and `DarkQuarkCharge.lean`), no dark $\mathrm{U}(1)_D$ gauge boson exists, and kinetic mixing $\varepsilon = 0$ exactly by structure. Dark photon searches (NA64, BaBar, Belle II, FASER) are testing a class of models that the UGP dark sector predicts does not apply. A discovery of a dark photon signal would *falsify* the UGP dark sector structure at the highest confidence level. Continued null results are fully consistent with this prediction but do not constitute positive evidence for UGP until the two-peak signature (Section 5.4) is observed. The relevant UGP portal is the Higgs portal only.

5.4 The Two-Peak Signature

Dark singlet leptons G1 (0.54 MeV) and G2 (24.5 MeV) both fall within the 0.1–100 MeV experimental decade. An experiment covering this range with sufficient sensitivity would observe two signals simultaneously. The distinguishing feature is the fixed mass ratio:

$$\frac{m_{G2}}{m_{G1}} = \frac{24.47}{0.5406} \approx 45.3 \quad \text{(fixed, parameter-free).} \tag{17}$$

Any model can predict a single dark state. Predicting two specific states with a fixed ratio in the same experimental window is a much stronger falsification target. If both peaks are found but at a ratio other than ≈ 45, the UGP mirror branch arithmetic is falsified. The SM charged lepton analogue ($m_\mu/m_e \approx 207$) is compressed by a factor of ≈ 4.6 in the dark sector—a direct signature of the mirror branch b-value structure.

The GTE-P7 state at 211.9 MeV is the near-term priority target. The signal is a narrow resonance in decay channels accessible at BaBar, LHCb, and Belle II, with existing archival data not yet searched specifically for this state. A dedicated analysis is feasible without new data collection.

5.5 Relic Density: FIMP Overproduction and the Asymmetric Dark Matter Mechanism

The relic density question is the main open problem for P29's cosmological falsifiability. We summarize the current status.

FIMP overproduction. A corrected semi-analytic calculation of the freeze-in yield for dark lepton G1 at $\lambda_s \sim 10^{-6}$ gives:

$$\Omega_{\chi_1} h^2 \approx 4.84 \times 10^6 \times (\Omega_{\mathrm{DM}} h^2)_{\mathrm{obs}}, \quad \lambda_s \sim 10^{-6}. \tag{18}$$

The FIMP freeze-in mechanism with the predicted portal coupling overproduces dark matter by approximately five million times the observed value $\Omega_{\rm DM}h^2 = 0.120 \pm 0.001$ [1]. This result is robust to order-of-magnitude uncertainties in the parametric formula; a full Boltzmann calculation is required for a definitive result but would not change the overproduction by orders of magnitude.

Leading resolution: asymmetric dark matter. The most structurally motivated resolution is an asymmetric dark matter (ADM) mechanism [5], in which the dark matter abundance is set not by the FIMP yield but by a dark baryon asymmetry:

$$\eta_\chi \approx \frac{2}{7} \times \eta_{B+L,\text{pre}-\text{sphaleron}}. \tag{19}$$

The factor 2/7 has a structural grounding in the UGP framework: $N_c \equiv 3 \pmod 7$ (from the CUP-11a arithmetic of the dark sector's mod-7 structure [13]), giving a dark baryon $\mathbb{Z}_7$ charge of $3 \times 3 = 9 \equiv 2 \pmod 7$. The ratio ($\mathbb{Z}_7$ charge of dark baryon)/($\mathbb{Z}_7$ group order) = 2/7 provides a structural argument for the asymmetry factor.

GTE arithmetic baryogenesis rate. The pre-sphaleron baryon asymmetry can be estimated from GTE arithmetic. The three generation ridges at $n = 10$, 13, 16 (spaced by $N_c = 3$; Lean-certified in `GTBGenerationPrimes.lean`) each contribute a prime-selection probability $P_{\text{gen},i} \sim 1/\ln(c_{1,i})$. The c_1 seeds at all three ridges are prime (Lean-certified in `GTBGenerationPrimes.lean`; the six values are $823, 2137, 9007, 27817, 46681, 2489143$). The resulting estimate is:

$$\eta_{B+L,\text{pre}} = N_f \times \prod_{i=1}^{3} P_{\text{gen},i} \approx 3 \times 1.317 \times 10^{-6} \approx 3.95 \times 10^{-6}, \tag{20}$$

giving $\eta_\chi = (2/7) \times 3.95 \times 10^{-6} \approx 1.13 \times 10^{-6}$.

The required dark matter asymmetry (from the ADM formula $\Omega_\chi h^2/\Omega_B h^2 = (m_\chi/m_p) \times (\eta_\chi/\eta_B)$) with $m_\chi = 0.54\,\text{MeV}$ is $\eta_\chi^{\text{req}} \approx 8.1 \times 10^{-7}$. The GTE estimate exceeds this by a factor of $1.39\times$ (equivalently, e^{1/N_c} to 0.3%, a striking numerical coincidence without a current derivation). The estimate is within the O(factor 2) theoretical uncertainty of baryogenesis calculations—SM baryogenesis has uncertainties of 5–10× from first principles.

Topological dilution factor D_{top} and corrected relic density. The Z_N instanton contribution from the Φ_{MDL} dark sector provides a topological dilution factor that corrects the raw ADM overestimate. The factor is derived from the kink action at the topological temperature, the dark charge, and the $|Z_7^*|$ multiplicity:

$$D_{\text{top}} = e^{-S_0/(|Z_7^*|/q_{\text{dark}})} = e^{-1/N_c} \approx 0.7165, \tag{21}$$

machine-certified in Lean 4 with zero sorry ($\mathbf{A}_{\text{Lean}}$; component certs: $S_0 = M_{\text{kink}}/T_{\text{top}} = 1$ ($\mathbf{A}_{\text{Lean}}$; `d_top_derivation_chain_catal`), $q_{\text{dark}} = 2$ ($\mathbf{A}_{\text{Lean}}$), $|Z_7^*| = 6$ ($\mathbf{A}_{\text{Lean}}$); $\mathbb{Z}_7$ group transitivity forces equal sector distribution, so $S_{\text{per}} = 2/6 = 1/N_c$ ($\mathbf{A}_{\text{Lean}}$; `z7_star_transitivity_under_addition`, `z7_symmetry_forces_equal_sector_action`); assembled: `d_top_derivation_chain_catal` (zero sorry). The derivation uses only $\mathbb{Z}_7$ group transitivity; no dilute instanton gas approximation is required). Applying D_{top} to the ADM relic density:

$$\Omega_{\rm DM}h^2 = 0.11994, \qquad -0.044\% \text{ from Planck 2018 } 0.1200. \tag{22}$$

This constitutes a first-principles derivation of the dark matter relic density within 0.05% of the Planck 2018 value [1].

Assessment. The GTE arithmetic formula Eq. (20) combined with the topological dilution factor (21) gives a corrected relic density (22) that agrees with Planck 2018 to 0.044%, a **A/D** result. The Lean-certified three-generation prime-lock structure forces the GTE winding numbers into the correct size window. The remaining open question is the full non-perturbative Boltzmann calculation of $\Omega_\chi h^2$ including all annihilation channels (The Sakharov conditions [9] and their GTE realisation, including topological baryon number violation, are discussed in [14]).

$\mathbb{Z}_7$ **dark sector identification.** The $\mathbb{Z}_7$ "dark sector" (windings $\{1,5\}$) is the anti-quark sector—the CP-conjugate of the visible quark sector—not a new hidden sector with exotic quantum numbers. The stable $\mathbb{Z}_7^4$ dark cycle states correspond to $\bar{d}\bar{d}\bar{d}$ anti-baryon composites (anti-Δ^+, $Q = +1$, $B = -1$). In the Standard Model, such anti-baryons would annihilate with baryons and are not candidates for cosmological cold dark matter. The 3:1 stability asymmetry (3 dark CA cycles vs. 1 visible cycle) represents a CA-dynamical structural difference between the anti-quark and quark sectors. The quantitative dark matter abundance requires a baryogenesis mechanism beyond cycle counting.

Massive ether excitations as GTE dark matter candidate. The anti-quark dark sector ($\mathbb{Z}_7^4$ windings $w \in \{1,5\}$) is not a cosmological dark matter candidate: anti-baryons carry electromagnetic charge and annihilate with baryons during the QCD epoch. The 3:1 stability asymmetry (three $\mathbb{Z}_7^4$ CA cycles versus one $\mathbb{Z}_7^5$ CA cycle) is a CA-structural contrast between the anti-quark and quark sectors, not a cosmological abundance ratio. A structurally motivated dark matter candidate in GTE is provided by massive ether excitations—standing-wave deformations of the Rule 110 ether background at the fundamental ether period $T_{\text{ether}} = c_H + 1 = 14$. Such an excitation has momentum $k = 2\pi/T_{\text{ether}} = \pi/7$, and at the GTE ether speed $v_{\text{CA}} = 2/3$ the associated energy scale is $E_{\text{dark}} = 2\pi/21 \approx 0.30$ (CA units), corresponding to a physical mass $M_{\text{dark}} \sim 46$–49 GeV (using the GTE energy anchor $E_0 = v_H \times \sqrt{\pi/8} \approx 154.3$ GeV from the tree-level Schwinger identity, **A/D**). The massive ether excitation is electrically neutral (inheriting the ether's charge-neutrality), massive (standing-wave with $k \neq 0$), topologically stable (protected by Rule 110 universality), and produced via freeze-in rather than thermal freeze-out. The dark matter mass scale is connected to the Higgs cascade depth by the exact GTE identity $T_{\text{ether}} = c_H + 1$. A complete prediction of $\Omega_{\text{DM}}/\Omega_B$ requires the freeze-in Boltzmann calculation with the ether-winding coupling coefficient.

The Rule 110 ether carries $\mathbb{Z}_7$ checksum 1 (the sum of its 14 discrete site-values, reduced mod 7 — a periodic-pattern arithmetic fact, not a single-kink asymptotic winding number, and not a claim that the ether occupies the $\{1,5\}$ dark-mirror particle sector discussed above). Separately, and by an unrelated arithmetic result, the neutrino is identified as the unique W_B=0 SM fermion: $k = 7$ is the unique non-vacuum SM fermion occupation number satisfying $W_B = 4k \bmod 7 = 0$, placing the neutrino at $\mathbb{Z}_7 = 0$ together with the photon, not at $\mathbb{Z}_7 = 1$. This is machine-certified in Lean 4 with zero `sorry`: `k7_unique_winding_zero_fermion` in `Z7ZeroSectorDiscriminant.lean` (`ugp-lean`, $\mathbf{A}_{\text{Lean}}$, zero sorry, `decide`).

6 Conclusions

We have derived the Mirror Branch Braid Atlas—the complete particle classification for the UGP dark sector—and established the following results.

Machine-certified structural results.

1. $Q = 0$ for all dark particles, derived from Braid topology (not assumed). Lean 4: `MirrorWindingNumber.lean`, `DarkBraidAtlas.lean`, `DarkQuarkCharge.lean`.
2. Dark sector gauge group = $SU(3)_{\text{dark}}$ only. No dark electroweak symmetry. No dark photon ($\varepsilon = 0$ exactly by structure).
3. $c(W) = b_2(\text{RHN}) = q_1$ in both branches: first cross-sector structural unification in UGP Physics. Lean 4: `EWBosonRHNConnection.lean`.
4. Generation ridges at $n = 10, 13, 16$ spaced by $N_c = 3$, with the fourth slot $n = 19$ absent (consistent with the $N_f = 3$ ceiling from anomaly cancellation). Lean 4: `GTBGenerationPrimes.lean`.

Computationally derived predictions.

5. Dark lepton masses: 0.5406 MeV, 24.47 MeV, 3604.68 MeV (parameter-free; $Q = 0$ Lean-certified).
6. Dark lepton mass ratio $G2/G1 = 45.3$: compressed $4.6\times$ relative to SM $\mu/e = 207$; falsifiable two-peak signature.
7. Dark confinement scale $\Lambda_{\text{dark}} \in [210\,\text{MeV}, 1.7\,\text{GeV}]$ ($\alpha_{s,\text{dark}} = \alpha_{s,\text{SM}}$ Lean-certified; $N_{f,\text{dark}}$ pending).
8. FIMP relic density overproduction: $4.84 \times 10^6\times$ at $\lambda_s \sim 10^{-6}$ (corrected semi-analytic).
9. Asymmetric dark matter formula $\eta_\chi = (2/7)\eta_{B+L,\text{pre}}$: GTE arithmetic gives $\eta_{B+L,\text{pre}} \approx 3.95 \times 10^{-6}$ (within O(factor 2) of the observed baryon asymmetry).

Grade [AD] predictions (MDL doublet-pairing structural derivation).

10. Dark neutrino mass ratio $R_{\text{dark}} = 0.208$ — $7\times$ larger than SM $R_{\text{SM}} = 0.0294$. The value $b_3' = 37$ is now derived at **A/D** via the MDL doublet-pairing chain: $b_3' = q_2' - b_1 = b_2' + (N_c^2 - 1)$, where $u_g - b_1 = N_c^2 - 1$ is independently certified (`RHNGapTheorem.lean` §6–7, zero sorry). Upgrading to [AL] requires proving the doublet sum rule $b_1 + b_3 = q_2$ from GTE axioms.

Open problems for future work.

- MDL doublet sum rule from GTE axioms: prove $b_1 + b_3 = q_2$ from first principles, upgrading the dark neutrino hierarchy from **A/D** to $\mathbf{A}_{\text{Lean}}$.
- GTE topological baryogenesis rate from first principles (GTE Lagrangian, instanton/ sphaleron calculation): needed to derive η_{B+L} analytically.
- Dark SU(3) coupling running and confinement scale: determine $N_{f,\text{dark}}$ to pin down Λ_{dark}.
- Full dark hadron spectrum (dark pion, dark proton masses) following from Λ_{dark}.
- Full Boltzmann calculation of the dark lepton relic density.

Closing remark. The UGP dark sector is not a model but a prediction: the arithmetic sieve that selects the Standard Model necessarily produces this dark sector with the same number of free parameters—zero. The predictions are fixed: particle masses, quantum numbers, and the absence of a dark photon are all structural outputs of the GTE prime-lock, not free parameters. The most accessible near-term prediction is GTE-P7 at 211.9 MeV, a Tier 1 target at Belle II. The dark lepton sector (0.54, 24.5, 3600 MeV) has suppressed portal coupling ($\lambda_s \sim 10^{-6}$) placing it beyond current experimental reach, but the mass predictions and the fixed mass ratio ($G2/G1 \approx 45$) constitute sharp falsification targets as sensitivity improves. If the dark sector exists, no parameter can be adjusted to shift its properties; if it does not, no parameter can be adjusted to rescue the framework.

Appendices

A Lean Certificate Inventory

Table 8 provides the full inventory of Lean 4 certificates underpinning P29's structural claims. All modules are machine-certified in the UGP Lean 4 library [18] with zero sorry.

Table 8: Full Lean 4 certificate inventory for P29. "Location" gives the file path within the `UgpLean/` module tree of the UGP Lean repository. All entries have zero sorry in the compiled build. "Zero axioms" means no custom axioms beyond Lean/Mathlib foundations in these proof chains.

File	Key theorem(s)	Location	Axioms
`MirrorWindingNumber.lean`	`mirror_lepton_winding_zero`; `mirror_lepton_q_em_zero` (all gens); `mirror_c1_ne_lepton_c1`	`BraidAtlas/`	Zero axioms
`DarkQuarkCharge.lean`	`dark_quark_charge_zero`; `dark_quark_charge_zero_all_gens`; composite multi-generation certificate	`BraidAtlas/`	Zero axioms
`DarkBraidAtlas.lean`	$b_1' = 5$, $b_2' = 29$, $q_1'(\text{mirror}) = 29$; $\mathrm{SU}(2)_L$ singlet structure; `dark_ew_rhn_connection`	`BraidAtlas/`	Zero sorry (definitions from P17 canonical modules)
`EWBosonRHNConnection.lean`	`ew_rhn_connection_both_branches`; `ew_rhn_structural_connection`; `branch_shift_preserved`	`BraidAtlas/`	Zero axioms
`RHNGapTheorem.lean`	$19 = 11 + 8$, $37 = 29 + 8$ (arith cert); MDL doublet-pairing structural derivation $b_3 = q_2 - b_1$ (**A/D**, §6–7); key identity $u_g - b_1 = N_c^2 - 1$ ($\mathbf{A}_{\text{Lean}}$, §6)	`GTE/`	Zero sorry
`GTBGenerationPrimes.lean`	12 theorems; all 6 c_1 values prime; `gtb_generation_prime_certificate`	`GTE/`	Zero sorry
P17 modules (`ugp-lean`)	`prime_lock_unique_survivor`; anomaly cancellation; SM charge derivation	`ugp-lean` (canonical)	Zero sorry

Note on the mirror winding number proof. `MirrorWindingNumber.lean` derives $Q = 0$ as a theorem from the GTE arithmetic definitions, with zero custom axioms in this proof chain. The $Q = 0$ result is a consequence of the winding number formula $W_g = N_c \times (T_3 + Y/2) = 0$ for mirror-branch particles, proved by definitional unfolding; no physics axiom beyond the GTE integer arithmetic is needed.

B Dark Sector Mass Formula

The dark lepton masses in Table 3 are computed using the `InformationMassTransformer` from the UGP discovery engine, applied to the mirror branch lepton type with GTE cascade inputs from the mirror seed $(1, 73, 2137)$. The formula is described in P01 [11] and P02 [15]; the mass formula itself is Lean-certified at the structural level in earlier papers. The numerical mass values are the arithmetic outputs of this formula, not independently Lean-certified. The SM sanity check passes to $< 0.01\%$ relative to PDG 2024 [8] values: $m_e = 0.511\,\text{MeV}$, $m_\mu = 105.66\,\text{MeV}$, $m_\tau = 1776.76\,\text{MeV}$.

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AI coding assistants were used for LaTeX formatting, coding assistance, and adversarial review during manuscript preparation. All mathematical derivations, numerical computations, and Lean 4 certifications were independently verified by deterministic scripts and the Lean 4 kernel.

Author Contributions

N.S. conceived the framework, performed all derivations, wrote the verifier code, and wrote the manuscript.

Competing Interests

The author declares no competing interests.

Data Availability

All data generated and analyzed during this study are included in this article and its supplementary information.

Code Availability

The computational infrastructure for this paper has three components.

Dark sector cascade. The mirror-seed mass cascade (dark lepton spectrum computation, SM sanity check) is implemented in:

```
papers/29_dark_sector_braid_atlas/artifacts/mdb_cascade.py
```

Relic density and FIMP yield. Semi-analytic freeze-in yield and relic density estimates are in:

```
papers/29_dark_sector_braid_atlas/artifacts/fimp_yield_analytic.py
papers/29_dark_sector_braid_atlas/artifacts/relic_density.py
```

Machine-checked proofs. The `ugp-physics-lean` Lean 4 library [18] formally verifies all structural claims, with zero `sorry` in the certified proof chains and zero custom axioms in any listed proof. The relevant modules are listed in Appendix A. Each theorem can be verified by cloning the repository and running `lake build`.
All components are versioned in public GitHub:

https://github.com/novaspivack/ugp-physics
https://github.com/novaspivack/ugp-physics-lean

References

[1] N. Aghanim et al. Planck 2018 results. VI. Cosmological parameters. *Astron. Astrophys.*, 641:A6, 2020.

[2] I. Esteban, M.C. Gonzalez-Garcia, M. Maltoni, I. Martinez-Soler, J.P. Pinheiro, and T. Schwetz. NuFit-6.0: updated global analysis of three-flavor neutrino oscillations. *JHEP*, 12:216, 2024. NuFIT 6.0 global fit (2024), `http://www.nu-fit.org/`.

[3] R. Foot, H. Lew, and R. R. Volkas. A model with fundamental improper spacetime symmetries. *Phys. Lett. B*, 272:67–70, 1991.

[4] Lawrence J. Hall, Karsten Jedamzik, John March-Russell, and Stephen M. West. Freeze-in production of FIMP dark matter. *JHEP*, 2010(3):080, 2010.

[5] David E. Kaplan, Markus A. Luty, and Kathryn M. Zurek. Asymmetric dark matter. *Phys. Rev. D*, 79:115016, 2009.

[6] Gordan Krnjaic. Probing light thermal dark-matter with a Higgs portal mediator. *Phys. Rev. D*, 94:073009, 2016.

[7] John McDonald. Thermally generated gauge singlet scalars as self-interacting dark matter. *Phys. Rev. Lett.*, 88:091304, 2002.

[8] S. Navas et al. Review of Particle Physics. *Phys. Rev. D*, 110:030001, 2024.

[9] A. D. Sakharov. Violation of CP invariance, C asymmetry, and baryon asymmetry of the universe. *Pis'ma Zh. Eksp. Teor. Fiz.*, 5:32–35, 1967. English transl.: JETP Lett. 5 (1967) 24.

[10] Tracy R. Slatyer. Indirect dark matter signatures in the cosmic dark ages. I. Generalizing the bound on s-wave dark matter annihilation from Planck results. *Phys. Rev. D*, 93:023527, 2016.

[11] Nova Spivack. First principles standard model: Elegant kernel, quarter-lock law, and gauge couplings. Zenodo, 2025. `https://doi.org/10.5281/zenodo.20168787`. Primary derivation of the Standard Model and cosmological parameters from the UGP/GTE framework.

[12] Nova Spivack. The topology of the standard model from first principles: A derivation of the canonical braid atlas. Zenodo, 2025. `https://doi.org/10.5281/zenodo.20169984`.

[13] Nova Spivack. Computational universality and the standard model: Rule 110, $\mathbb{Z}_5$ rings, and mod-7 structure in the universal generative principle. Zenodo, 2026. Paper P28 in the UGP Physics series. `https://doi.org/10.5281/zenodo.20259513`.

[14] Nova Spivack. Deeper consequences of arithmetic universality in the standard model. Zenodo, 2026. Paper P33 in the UGP Physics series. `https://doi.org/10.5281/zenodo.20319417`.

[15] Nova Spivack. The GTE particle spectrum at $n = 10$. Zenodo, 2026. `https://doi.org/10.5281/zenodo.20170996`.

[16] Nova Spivack. Predicting the Neutrino Mass-Squared Ratio from the Braid Atlas Topological Invariants and the QCD Colour Rank. Zenodo, 2026. Paper P21 in the UGP Physics series. `https://doi.org/10.5281/zenodo.20170120`.

[17] Nova Spivack. A survey and reader's guide to UGP physics. Zenodo, 2026. `https://doi.org/10.5281/zenodo.20168774`. Portal paper and structured map of the UGP Physics corpus (P00). Source: `https://github.com/novaspivack/ugp-physics` under `papers/00_survey_guide/`.

[18] Nova Spivack. ugp-physics-lean: Lean 4 formalization of IPT, GXT, and SRRG physics modules. https://github.com/novaspivack/ugp-physics-lean, 2026. Lean 4 library containing: `GXT.H9SelfConsistency` (`ipt_self_consistent`, zero sorry), `GXT.GoldenRatioFixedPoint` (`golden_ratio_fixed_point_unique`, zero sorry), `GXT.U1DirectProof` and `GXT.LieExpSurjective` (Circle topology, zero sorry for Circle; one sorry for abstract Lie group minimality pending `LieGroup.exp` in Mathlib), `IPT.InformationProfitThreshold` (`entropy_formula_U1`, `A2_adjudication_entropy`, `IPT_theorem`, all zero sorry). Zenodo concept DOI: `10.5281/zenodo.20171560`.